Forecasting the 10-Year U.S. Treasury Yield Using Yield-Curve Factors and AutoReg Models

10-observation-ahead forecasting and expanding-window backtesting

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Executive summary

This project compares a four-factor OLS model with an AR(10) model for forecasting the 10-year U.S. Treasury yield 10 available observations ahead. The latest factor forecast is 5.0322% from a signal date of 19 October 2026, with an estimated weekday target date of 2 November 2026. In expanding-window backtests, neither model beats the unchanged-yield benchmark: the factor model records an out-of-sample R-squared of -31.1357%, while AR(10) records -67.0541%. The factor model is less inaccurate than AR(10), but the persistence benchmark is preferred for this horizon.

1. Objective and data

The project tests whether information from the U.S. Treasury yield curve can improve forecasts of DGS10, the 10-year constant-maturity Treasury yield. The factor model uses current DGS10, DGS5, DGS2 and DGS1MO yields, while the competing AR(10) model uses only the history of DGS10.

- Forecast horizon: 10 available FRED observations ahead.
- Selected sample: 2 January 2024 to 19 October 2026.
- Cleaned full dataset: 6,306 observations from 31 July 2001 to 19 October 2026.
- Backtest: expanding-window forecasts over the final 20% of usable observations.
- Benchmark: future DGS10 equals DGS10 on the signal date.

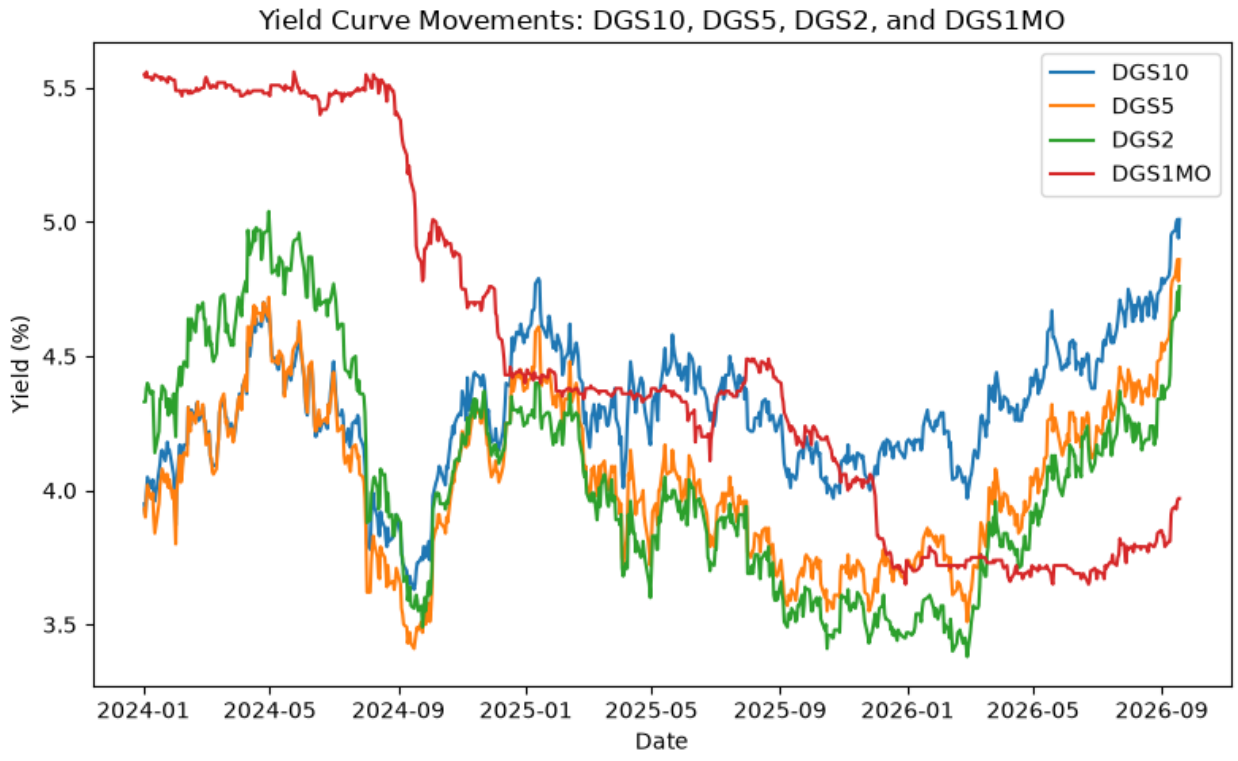


Figure 1. U.S. Treasury yield movements in the selected sample.

2. Methodology

Four-factor OLS model

DGS10 is shifted 10 observations forward to create the forecast target. Current DGS10, DGS5, DGS2 and DGS1MO are used as explanatory variables. Residual tests identify both serial correlation and heteroskedasticity, so HAC standard errors with a maximum lag of 20 are used for coefficient inference. HAC changes standard errors, p-values and confidence intervals, but not the fitted coefficients or point forecasts.

AR(10) model

The univariate model uses 10 lags of DGS10. AIC selects lags 1 through 5 and BIC selects lag 1, but AR(10) is retained to preserve the project specification and provide a richer lag comparison.

Evaluation measures

Forecast performance is measured with MAE, RMSE and benchmark-relative out-of-sample R-squared. Positive OOS R-squared indicates that a model beats the unchanged-yield benchmark; a negative value indicates inferior performance.

3. Four-factor model results

Variable	Coefficient	HAC p-value	Interpretation
Constant	2.0564	< 0.001	Positive intercept.
DGS10	0.2438	0.304	Not significant after HAC adjustment.
DGS5	0.3579	0.337	Not significant after HAC adjustment.
DGS2	0.1554	0.459	Not significant after HAC adjustment.
DGS1MO	-0.1920	< 0.001	Significant negative predictor after HAC adjustment.

Model fit: R-squared = 0.753; adjusted R-squared = 0.752; 689 observations.

Inference: Only DGS1MO remains statistically significant after the HAC adjustment. Strong co-movement among Treasury maturities and persistent residuals increase coefficient uncertainty.

4. Latest forecast and probability analysis

Item	Result
Signal date	19 October 2026
Estimated weekday target date	2 November 2026
Latest inputs	DGS10 5.15%; DGS5 4.94%; DGS2 4.77%; DGS1MO 4.11%
Factor-model forecast	DGS10 = 5.0322%

Normal P(DGS10 > 4.955%)	71.1220%
Student t P(DGS10 > 4.955%)	71.1914%

The estimated target date excludes weekends but does not explicitly adjust for U.S. market holidays. The model horizon is therefore most accurately described as 10 available observations ahead. Both distributional assumptions imply approximately a 71% probability that DGS10 exceeds 4.955% at the horizon.

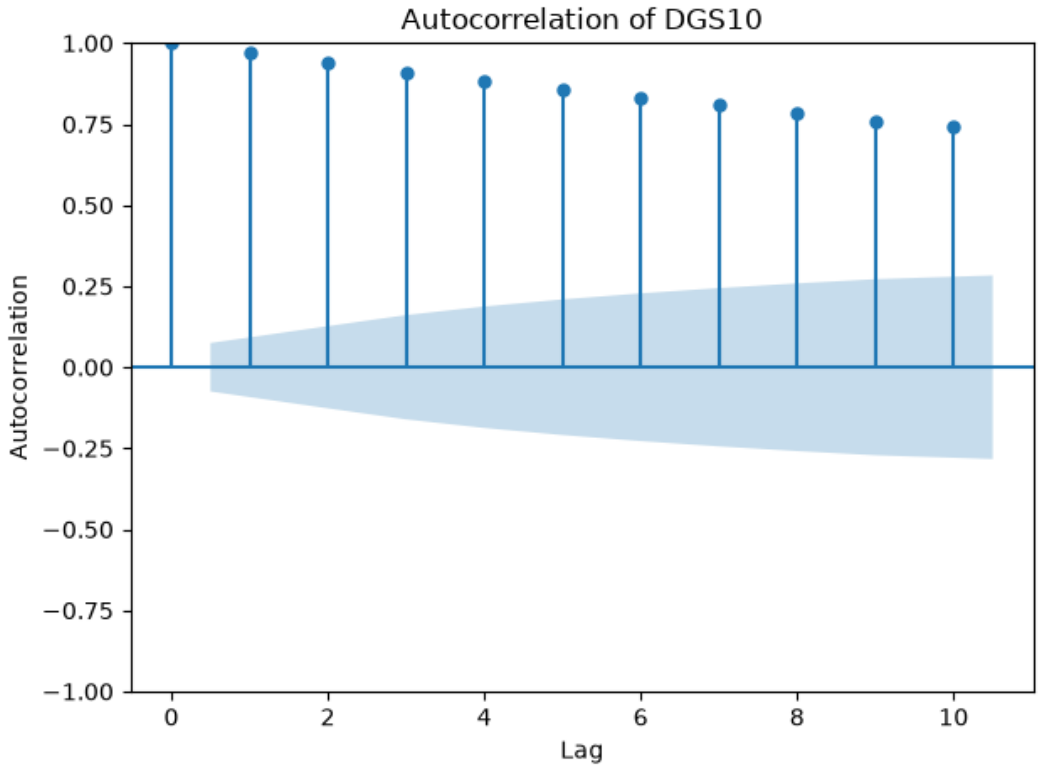


Figure 2. DGS10 autocorrelation through lag 10.

5. Diagnostics

Diagnostic	Result	Implication
OLS Ljung-Box, lag 10	p-value = 0.000	Very strong residual serial correlation.
White test	p-value ≈ 0.000	Strong evidence of heteroskedasticity.
HAC covariance	20 lags	Used for robust coefficient inference.
AR(10) Ljung-Box, lag 10	p-value = 0.999665	No evidence of remaining autocorrelation through lag 10.

The ACF chart shows very high persistence in DGS10, with autocorrelation remaining positive and large through lag 10. This persistence explains why the unchanged-yield benchmark is difficult to beat. Although AR(10) removes residual autocorrelation, clean residuals alone do not guarantee accurate forecasts.

6. Out-of-sample backtest

Model	MAE	RMSE	Benchmark MAE	Benchmark RMSE	OOS R^2
Four-factor OLS	0.114491	0.151382	0.102754	0.132194	-31.1357%
AR(10)	0.139507	0.174326	0.105154	0.134876	-67.0541%

The four-factor model has higher MAE and RMSE than the unchanged-yield benchmark. Its OOS R-squared of -31.1357% means its forecast mean squared error is approximately 31.1% higher than the benchmark error. The AR(10) result is weaker: its OOS R-squared of -67.0541% indicates an approximately 67.1% increase in mean squared error relative to persistence.

Among the two fitted models, the factor model performs better than AR(10), but the benchmark remains the preferred forecasting rule for this sample and 10-observation horizon.

7. Model decision

Criterion	Four-factor OLS	AR(10)	Persistence benchmark
Uses curve information	Yes	No	No
Residual autocorrelation	Strong	Not detected	Not applicable
OOS R-squared	-31.1357%	-67.0541%	0% by definition
Preferred for forecasting	No	No	Yes

8. Limitations and next steps

- The 10-observation horizon is longer than the earlier six-observation horizon and is more difficult to forecast; model performance deteriorates materially.
- Treasury yields are highly persistent and the factor variables are strongly correlated, increasing multicollinearity and coefficient uncertainty.
- HAC corrects inference but does not remove serial correlation or improve point forecasts.
- The selected sample begins in 2024, so results may be sensitive to the interest-rate regime.
- Future work should compare AR(1), the AIC-selected lag set, yield changes rather than yield levels, regularized regression, and rolling-window estimation.

9. Conclusion

The latest four-factor model forecasts DGS10 at 5.0322% 10 available observations after 19 October 2026. However, the backtest shows that both the factor model and AR(10) underperform a simple unchanged-yield forecast. The factor model is superior to AR(10) but still has a negative OOS R-squared of -31.1357%. The appropriate project conclusion is therefore not that the factor model is predictively successful, but that Treasury-yield persistence makes the benchmark difficult to beat and that model evaluation must prioritize out-of-sample results over in-sample fit.